

Exercise #8.2

In each of Problem 1 – 4, find parametric equations, direction ratios, direction cosines and measures of the direction angles of the straight line through P and Q:

Q#1: $P(1, -2, 0)$, $Q(5, -10, 1)$

Solution:

Given points are $P(1, -2, 0)$ and $Q(5, -10, 1)$

Let $R(x, y, z)$ is another point on the straight line then equation of straight line will be written as

$$\frac{x-1}{5-1} = \frac{y+2}{-10+2} = \frac{z-0}{1-0} \Rightarrow \frac{x-1}{4} = \frac{y+2}{-8} = \frac{z}{1} = t$$

Parametric equations of given straight line can be written as

$$\frac{x-1}{4} = t \Rightarrow x = 1 + 4t$$

$$\frac{y+2}{-8} = t \Rightarrow y = -2 - 8t$$

$$\frac{z}{1} = t \Rightarrow z = 0 + 1t$$

$\overline{\hspace{10em}}$
 $P(1, -2, 0) \qquad Q(5, -10, 1) \qquad R(x, y, z)$

Direction ratios of straight line PQ are 4, -8, 1.

Let $\vec{d}$ is the direction vector , $\vec{d} = 4\hat{i} - 8\hat{j} + \hat{k} \Rightarrow |\vec{d}| = \sqrt{4^2 + (-8)^2 + 1^2} = \sqrt{81} = 9$

Direction cosines of line through P & Q are

$$\cos \alpha = \frac{4}{9}, \cos \beta = \frac{-8}{9}, \cos \gamma = \frac{1}{9}$$

Directional angles of line PQ are

$$\alpha = \cos^{-1} \frac{4}{9}, \beta = \cos^{-1} \frac{-8}{9}, \gamma = \cos^{-1} \frac{1}{9}$$

$$\Rightarrow \alpha = 63^\circ 37', \beta = 152^\circ 44', \gamma = 83^\circ 37'$$

Q#2: $P(6, 5, -3)$, $Q(4, 1, 1)$

Now do yourself as above

Q#3: $P(1, -5, 1)$, $Q(4, -5, 4)$

Do yourself as above

Q#4: P(3, 5, 7), Q(6, -8, 10)

Solution:

Given points are P(3,5,7) & Q(6, -8, 10)

Let R(x, y, z) is another point on the straight line then equation of straight line will be written as

$$\frac{x-3}{6-3} = \frac{y-5}{-8-5} = \frac{z-7}{10-7}$$

$$\Rightarrow \frac{x-3}{3} = \frac{y-5}{-13} = \frac{z-7}{3} = t$$

Parametric equations of given straight line can be written as

$$\frac{x-3}{3} = t \Rightarrow x = 3 + 3t$$

$$\frac{y-5}{-13} = t \Rightarrow y = 5 - 13t$$

$$\frac{z-7}{3} = t \Rightarrow z = 7 + 3t$$

Direction ratios of straight line PQ are 3, -13, 3.

P(3,5,7) Q(6,-8,10) R(x,y,z)

Let $\vec{d}$ is the direction vector

$$\vec{d} = 3\hat{i} - 13\hat{j} + 3\hat{k} \Rightarrow |\vec{d}| = \sqrt{3^2 + (-13)^2 + 3^2} = \sqrt{187}$$

Direction cosines of line through P & Q are

$$\cos \alpha = \frac{3}{\sqrt{187}}, \cos \beta = \frac{-13}{\sqrt{187}}, \cos \gamma = \frac{3}{\sqrt{187}}$$

Directional angles of line PQ are

$$\alpha = \cos^{-1} \frac{3}{\sqrt{187}}, \beta = \cos^{-1} \frac{-13}{\sqrt{187}}, \gamma = \cos^{-1} \frac{3}{\sqrt{187}}$$

$$\Rightarrow \alpha = 77^\circ 19', \beta = 159^\circ 19', \gamma = 77^\circ 19'$$

Q#5: Find the direction cosines the coordinate axis.

Solution:

We want to find the direction cosine of x-axis, y-axis and z-axis.

(I) Let x-axis makes angles $0^\circ, 90^\circ, 90^\circ$ with x-axis, y-axis and z-axis

So direction cosines of x-axis are

$$\cos 0^\circ = 1, \cos 90^\circ = 0, \cos 90^\circ = 0$$

(II) Let y-axis makes angles $90^\circ, 0^\circ, 90^\circ$ with x-axis, y-axis and z-axis

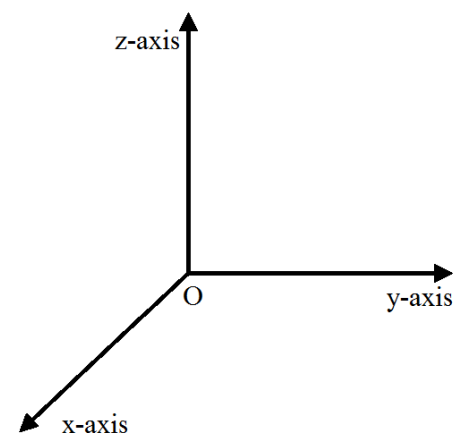
So direction cosines of y-axis are

$$\cos 90^\circ = 0, \cos 0^\circ = 1, \cos 90^\circ = 0$$

(III) Let z-axis makes angles $90^\circ, 90^\circ, 0^\circ$ with x-axis, y-axis and z-axis

So direction cosines of z-axis are

$$\cos 90^\circ = 0, \cos 90^\circ = 0, \cos 0^\circ = 1$$

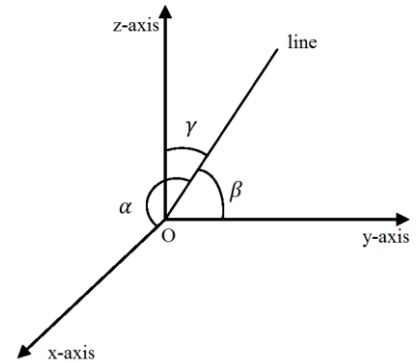


Q#6: Prove that if measures of the direction angles of a straight line are α, β and γ , then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$.

Solution: Let α, β and γ be the directional angles of a straight line then direction cosines of line are $\cos \alpha, \cos \beta$ & $\cos \gamma$.

As we know that

$$\begin{aligned}\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma &= 1 \\ 1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma &= 1 \\ 3 - (\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma) &= 1 \\ 3 - 1 &= (\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma) \\ \Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma &= 2 \quad \text{Hence proved.}\end{aligned}$$



Q#7: If measures of two of the direction angles of a straight line are 45° and 60° , find measure of the third direction angle.

Solution: Let α, β and γ be the directional angles of a straight line $\alpha = 45^\circ, \beta = 60^\circ, \gamma = ?$

As we know that

$$\begin{aligned}\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma &= 1 \\ \cos^2 45^\circ + \cos^2 60^\circ + \cos^2 \gamma &= 1 \\ \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \cos^2 \gamma &= 1 \quad \Rightarrow \frac{1}{2} + \frac{1}{4} + \cos^2 \gamma = 1 \\ \cos^2 \gamma &= 1 - \frac{3}{4} \quad \Rightarrow \cos^2 \gamma = \frac{1}{4} \quad \Rightarrow \cos \gamma = \frac{1}{\sqrt{4}} \quad \Rightarrow \cos \gamma = \frac{1}{2} \quad \Rightarrow \gamma = \cos^{-1}\left(\frac{1}{2}\right) \\ \Rightarrow \gamma &= 60^\circ \quad \text{is required direction angle.}\end{aligned}$$

Q#8: The direction cosines l, m, n of two straight lines are given by the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$. Find measure of the angle between them.

Solution: We have to find the angle between two straight lines.

Let l_1, m_1, n_1 be the direction cosines of line L and l_2, m_2, n_2 be the direction cosines of line M.

Let θ be the angle between line L and M then it is described as

$$\begin{aligned}\cos \theta &= l_1 l_2 + m_1 m_2 + n_1 n_2 \quad \text{----- (A)} \\ l + m + n &= 0 \quad \text{----- (1)} \\ l^2 + m^2 - n^2 &= 0 \quad \text{----- (2)} \\ (1) \Rightarrow n &= -(l + m) \quad \text{----- (3)}\end{aligned}$$

Putting in eq.(2)

$$\begin{aligned}l^2 + m^2 - [-(l + m)]^2 &= 0 \\ l^2 + m^2 - [l^2 + m^2 + 2lm] &= 0 \\ l^2 + m^2 - l^2 - m^2 - 2lm &= 0 \\ -2lm &= 0 \quad \Rightarrow lm = 0\end{aligned}$$

Either $l = 0$ or $m = 0$

Put $l = 0$ in eq. (3)

$$\Rightarrow n = -\frac{m}{1}$$

$$\Rightarrow \frac{n}{1} = \frac{-1}{1}$$

$$\frac{l}{0} = \frac{n}{1} = \frac{m}{-1} \Rightarrow \frac{l^2 + m^2 + n^2}{\sqrt{0^2 + 1^2 + (-1)^2}} = \frac{1}{\sqrt{2}}$$

So direction cosines of line L are

$$l_1 = 0, \quad m_1 = \frac{-1}{\sqrt{2}}, \quad n_1 = \frac{1}{\sqrt{2}}$$

Put $m = 0$ in eq. (3)

$$\Rightarrow n = -\frac{l}{1}$$

$$\Rightarrow \frac{n}{1} = \frac{-1}{1}$$

$$\frac{l}{-1} = \frac{n}{1} = \frac{m}{0} \Rightarrow \frac{l^2 + m^2 + n^2}{\sqrt{(-1)^2 + 1^2 + 0^2}} = \frac{1}{\sqrt{2}}$$

So direction cosines of line M are

$$l_2 = \frac{-1}{\sqrt{2}}, \quad m_2 = 0, \quad n_2 = \frac{1}{\sqrt{2}}$$

Using in equation (A)

$$\cos \theta = (0) \left(\frac{-1}{\sqrt{2}} \right) + \left(\frac{-1}{\sqrt{2}} \right) (0) + \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) \Rightarrow \cos \theta = 0 + 0 + \frac{1}{2} \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \cos^{-1} \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

Q#9: The direction cosines l, m, n of two straight lines are given by the equations $l + m + n = 0$ and $2lm + 2ln - mn = 0$. Find measure of the angle between them.

Solution: We have to find the angle between two straight lines.

Let l_1, m_1, n_1 be the direction cosines of line L and l_2, m_2, n_2 be the direction cosines of line M.

Let θ be the angle between line L and M then it is described as $\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$ ----- (A)

$$l + m + n = 0 \text{ ----- (1)}$$

$$2lm + 2ln - mn = 0 \text{ ----- (2)}$$

$$(1) \Rightarrow n = -(l + m) \text{ ----- (3)}$$

Putting in eq. (2) $2lm + 2l[-(l + m)] - m[-(l + m)] = 0$

$$2lm - 2l(l + m) + m(l + m) = 0$$

$$2lm - 2l^2 - 2lm + lm + m^2 = 0$$

$$-2l^2 + lm + m^2 = 0$$

$$2l^2 - lm - m^2 = 0$$

$$2l^2 - 2lm + lm - m^2 = 0$$

$$2l(l - m) + m(l - m) = 0 \Rightarrow (l - m)(2l + m) = 0$$

$$l - m = 0$$

$$\text{or } l = m \Rightarrow \frac{l}{1} = \frac{m}{1}$$

Putting in eq. (3)

$$n = -2l$$

$$\frac{n}{-2} = \frac{l}{1}$$

$$\text{So } \frac{l}{1} = \frac{m}{1} = \frac{n}{-2} \Rightarrow \frac{l^2 + m^2 + n^2}{\sqrt{1^2 + 1^2 + (-2)^2}} = \frac{1}{\sqrt{6}}$$

So direction cosines of line L are

$$l_1 = \frac{1}{\sqrt{6}}, \quad m_1 = \frac{1}{\sqrt{6}}, \quad n_1 = \frac{-2}{\sqrt{6}}$$

Using in equation (A)

$$\cos \theta = \left(\frac{1}{\sqrt{6}} \right) \left(\frac{1}{\sqrt{6}} \right) + \left(\frac{1}{\sqrt{6}} \right) \left(\frac{-2}{\sqrt{6}} \right) + \left(\frac{-2}{\sqrt{6}} \right) \left(\frac{1}{\sqrt{6}} \right) = \frac{1}{6} - \frac{2}{6} - \frac{2}{6} = \frac{1-2-2}{6} = -\frac{3}{6} \Rightarrow \cos \theta = -\frac{1}{2}$$

$$\Rightarrow \theta = \cos^{-1} \left(-\frac{1}{2} \right) \Rightarrow \theta = 60^\circ$$

$$2l + m = 0$$

$$\text{or } m = -2l \Rightarrow \frac{l}{1} = \frac{m}{-2}$$

Putting in eq. (3)

$$n = l$$

$$\frac{l}{1} = \frac{n}{1}$$

$$\text{So } \frac{l}{1} = \frac{m}{-2} = \frac{n}{1} \Rightarrow \frac{l^2 + m^2 + n^2}{\sqrt{1^2 + 1^2 + (-2)^2}} = \frac{1}{\sqrt{6}}$$

So direction cosines of line M are

$$l_2 = \frac{1}{\sqrt{6}}, \quad m_2 = \frac{-2}{\sqrt{6}}, \quad n_2 = \frac{1}{\sqrt{6}}$$

Find equations of the straight line L and M in symmetric forms. Determine whether the pairs of lines intersect. Find the point of intersection if it exists.

Q#10: L : through A(2, 1, 3), B(-1, 2, -4)
M : through P(5, 1, -2), Q(0, 4, 3)

Solution:

The equation of the straight line L through A(2, 1, 3) & B(-1, 2, -4)

$$\frac{x-2}{2-1} = \frac{y-1}{1-2} = \frac{z-3}{3-4} \Rightarrow \frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-3}{7}$$

The equation of the straight line M through P(5, 1, -2) & Q(0, 4, 3)

$$\frac{x-5}{5-0} = \frac{y-1}{1-4} = \frac{z+2}{-2-3} \Rightarrow \frac{x-5}{5} = \frac{y-1}{-3} = \frac{z+2}{-5}$$

Which are the required equations in symmetric form of L & M.

Now we write the equations of L & M in parametric forms.

$$\text{Let } \frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-3}{7} = t \quad (\text{say})$$

$$\text{let } \frac{x-5}{5} = \frac{y-1}{-3} = \frac{z+2}{-5} = s \quad (\text{say})$$

Now parametric equations of lines L & M are

$$\begin{aligned} \frac{x-2}{3} = t & \Rightarrow x = 2 + 3t \\ \frac{y-1}{-1} = t & \Rightarrow y = 1 - t \\ \frac{z-3}{7} = t & \Rightarrow z = 3 + 7t \end{aligned} \Rightarrow L : \begin{cases} x = 2 + 3t \\ y = 1 - t \\ z = 3 + 7t \end{cases}$$

$$\begin{aligned} \frac{x-5}{5} = s & \Rightarrow x = 5 + 5s \\ \frac{y-1}{-3} = s & \Rightarrow y = 1 - 3s \\ \frac{z+2}{-5} = s & \Rightarrow z = -2 - 5s \end{aligned} \Rightarrow M : \begin{cases} x = 5 + 5s \\ y = 1 - 3s \\ z = -2 - 5s \end{cases}$$

Let the lines L & M intersect at P(x, y, z) so this point will lie on both lines L & M.

Comparing above equations

$$\Rightarrow 2 + 3t = 5 + 5s \Rightarrow 3t - 5s = 3 \quad \text{----- (1)}$$

$$1 - t = 1 - 3s \Rightarrow t - 3s = 0 \quad \text{----- (2)}$$

$$3 + 7t = -2 - 5s \Rightarrow 7t + 5s = -5 \quad \text{----- (3)}$$

Now multiply eq. (2) by 3 and subtracting

$$4s = 3 \Rightarrow s = \frac{3}{4}$$

Put in eq. (2)

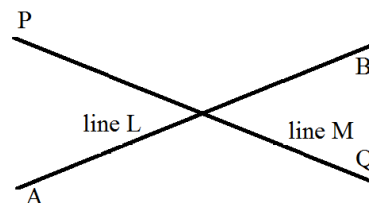
$$t - 3\left(\frac{3}{4}\right) = 0 \Rightarrow t = \frac{9}{4}$$

Now putting these values in equation (3) we have

$$7\left(\frac{9}{4}\right) + 5\left(\frac{3}{4}\right) = -5 \Rightarrow \frac{63}{4} + \frac{15}{4} = -5 \Rightarrow \frac{78}{4} \neq -5$$

We see that these values of t & s do not satisfy equation (3)

Hence, given straight lines L & M do not intersect. So point of intersection doesn't exist.



Q#11: L : $\vec{r} = (3\hat{i} + 2\hat{j} - \hat{k}) + t(6\hat{i} - 4\hat{j} - 3\hat{k})$

M : $\vec{r} = (5\hat{i} + 4\hat{j} + 7\hat{k}) + s(14\hat{i} - 6\hat{j} + 2\hat{k})$

Solution:

For given straight line L

$$\vec{r} = (3 + 6t)\hat{i} + (2 - 4t)\hat{j} + (-1 - 3t)\hat{k}$$

For given straight line M

$$\vec{r} = (5 + 14s)\hat{i} + (4 - 6s)\hat{j} + (-7 + 2s)\hat{k}$$

Now parametric equations for line L are

$$\left. \begin{array}{l} x = 3 + 6t \\ y = 2 - 4t \\ z = -1 - 3t \end{array} \right\} \text{-----(A)}$$

Parametric equations for given line M are

$$\left. \begin{array}{l} x = 5 + 14s \\ y = 4 - 6s \\ z = 7 + 2s \end{array} \right\} \text{-----(B)}$$

Now equations of straight lines L & M are in symmetric form

$$\frac{x-3}{6} = \frac{y-2}{-4} = \frac{z+1}{-3} \text{ -----(L)}$$

$$\frac{x-5}{14} = \frac{y-4}{-6} = \frac{z-7}{2} \text{ -----(M)}$$

Let the lines L & M intersect at (x, y, z) so this point will lie on both lines L & M.

Comparing equations (A) & (B)

$$\Rightarrow 3 + 6t = 5 + 14s \Rightarrow 6t - 14s = 2 \Rightarrow 3t - 7s = 1 \text{ -----(1)}$$

$$2 - 4t = 4 - 6s \Rightarrow 4t - 6s = -2 \text{ -----(2)}$$

$$-1 - 3t = 7 + 2s \Rightarrow 3t + 2s = -8 \text{ -----(3)}$$

Subtracting eq. (1) & (3)

$$-9s = 9 \Rightarrow s = -1$$

Putting value of s in equation (1)

$$3t - 7(-1) = 1 \Rightarrow 3t = 1 - 7 \Rightarrow 3t = -6 \Rightarrow t = -2$$

Now putting values of s & t in equation (2)

$$4(-2) - 6(-1) = -2 \Rightarrow -8 + 6 = -2 \Rightarrow -2 = -2$$

We see that these values of s & t satisfy equation (2)

Hence given straight lines L & M intersect.

For point of intersection we put $s = -1$ in equation (B)

Hence point of intersection of given straight line is $(x, y, z) = (-9, 10, 5)$.

Q#12: L : through A(2, -1, 0) and parallel to $\vec{b} = [4, 3, 2]$

M : through P(-1, 3, 5) and parallel to $\vec{c} = [1, 7, 3]$

Solution:

Symmetric form of given straight line L through the point A(2, -1, 0) and parallel to $\vec{b} = [4, 3, 2]$

$$\frac{x-2}{4} = \frac{y+1}{3} = \frac{z-0}{2}$$

let $\frac{x-2}{4} = \frac{y+1}{3} = \frac{z}{-2} = t$

Symmetric form of given straight line M through the point P(-1, 3, 5) and parallel to $\vec{c} = [1, 7, 3]$.

$$\frac{x+1}{1} = \frac{y-3}{7} = \frac{z-5}{3}$$

let $\frac{x+1}{1} = \frac{y-3}{7} = \frac{z-5}{3} = s$

Now parametric equations of lines L & M are

for line L $\frac{x-2}{4} = t \Rightarrow x = 2 + 4t$
 $\frac{y+1}{3} = t \Rightarrow y = -1 + 3t$
 $\frac{z}{-2} = t \Rightarrow z = 0 - 2t$

for line M $\frac{x+1}{1} = s \Rightarrow x = -1 + 1s$
 $\frac{y-3}{7} = s \Rightarrow y = 3 + 7s$
 $\frac{z-5}{3} = s \Rightarrow z = 5 + 3s$

NOW DO YOURSELF AS ABOVE

- Find the distance of the given point P from the given line L.

Q#13: P = (3, -2, 1), L : $\begin{cases} x = 1 + t \\ y = 3 - 2t \\ z = -2 + 2t \end{cases}$

Solution: Given point and given line are

$P = (3, -2, 1), \quad L : \begin{cases} x = 1 + t \\ y = 3 - 2t \\ z = -2 + 2t \end{cases}$

A point on the line L is A = (1, 3, -2)

Now $\vec{AP} = P(3, -2, 1) - A(1, 3, -2) = [3 - 1, -2 - 3, 1 + 2]$

$$\vec{AP} = 2\hat{i} - 5\hat{j} + 3\hat{k}$$

Now direction vector of the given line L is $\vec{b} = 1\hat{i} - 2\hat{j} + 2\hat{k}$

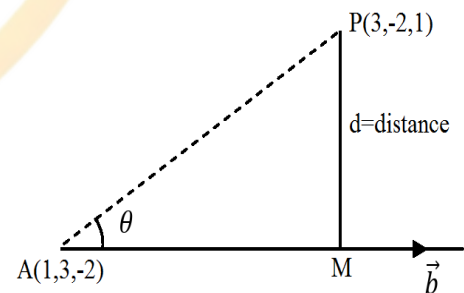
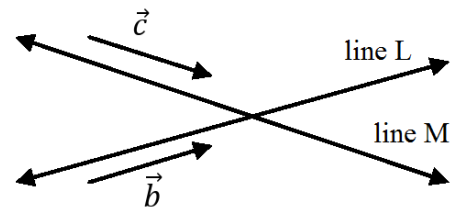
Let d be the required distance of a point from a line L then by using formula

$$d = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|} \quad \text{--- (A)}$$

$$\therefore \vec{AP} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -5 & 3 \\ 1 & -2 & 2 \end{vmatrix} = (-10 + 6)\hat{i} - (4 - 3)\hat{j} + (-4 + 5)\hat{k} \Rightarrow \vec{AP} \times \vec{b} = -4\hat{i} - \hat{j} + \hat{k}$$

$$|\vec{AP} \times \vec{b}| = \sqrt{(-4)^2 + (-1)^2 + 1^2} = \sqrt{16 + 1 + 1} = \sqrt{18} \quad \& \quad |\vec{b}| = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

Putting in equation (A) $d = \frac{\sqrt{18}}{3}$ is required distance



Q#14: $P = (0, -2, 1)$, $L : \frac{x-1}{4} = \frac{y+3}{-2} = \frac{z+1}{5}$

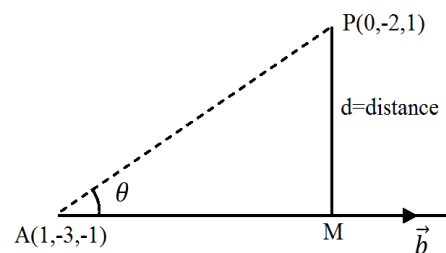
Solution: A point on the line L is $A = (1, -3, -1)$

$$\overrightarrow{AP} = P(0, -2, 1) - A(1, -3, -1) = [0 - 1, -2 + 3, 1 + 1]$$

$$\overrightarrow{AP} = -\hat{i} + \hat{j} + 2\hat{k}$$

Now direction vector of the given line L is $\vec{b} = 4\hat{i} - 2\hat{j} + 5\hat{k}$

DO YOURSELF AS ABOVE



Q#15: If the edges of a rectangular parallelepiped are a, b, c ; show that the angles between the four diagonals are given by

$$\arccos\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}\right)$$

Solution:

Consider a parallelepiped as shown in the figure.

Here the lengths of the edges OA, OB & OC are a, b, c respectively.

The coordinates of the vertices of parallelepiped are

$$O = (0, 0, 0), O' = (a, b, c), A = (a, 0, 0), A' = (0, b, c)$$

$$B = (0, b, 0), B' = (a, 0, c), C = (0, 0, c), C' = (a, b, 0)$$

The four diagonals of given parallelepiped are

$$\overrightarrow{OO'}, \overrightarrow{AA'}, \overrightarrow{BB'}, \overrightarrow{CC'}$$

Now

$$\overrightarrow{OO'} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\overrightarrow{AA'} = -a\hat{i} + b\hat{j} + c\hat{k}$$

$$\overrightarrow{BB'} = a\hat{i} - b\hat{j} + c\hat{k}$$

$$\overrightarrow{CC'} = a\hat{i} + b\hat{j} - c\hat{k}$$

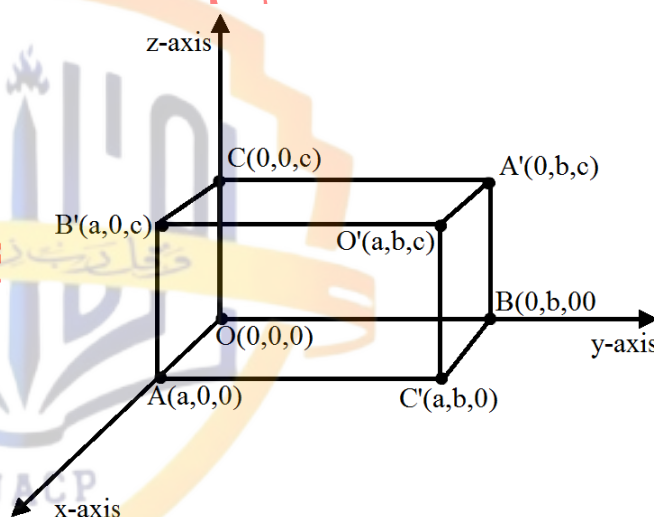
So direction cosines of

$$\overrightarrow{OO'} \Rightarrow l_1 = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m_1 = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n_1 = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$$\overrightarrow{AA'} \Rightarrow l_2 = \frac{-a}{\sqrt{a^2 + b^2 + c^2}}, m_2 = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n_2 = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$$\overrightarrow{BB'} \Rightarrow l_3 = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m_3 = \frac{-b}{\sqrt{a^2 + b^2 + c^2}}, n_3 = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$$\overrightarrow{CC'} \Rightarrow l_4 = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m_4 = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n_4 = \frac{-c}{\sqrt{a^2 + b^2 + c^2}}$$



Let α be the angle between $\overrightarrow{OO'}$ and $\overrightarrow{AA'}$

$$\cos \alpha = l_1 l_2 + m_1 m_2 + n_1 n_2$$

$$= \left(\frac{a}{\sqrt{a^2+b^2+c^2}} \right) \left(\frac{-a}{\sqrt{a^2+b^2+c^2}} \right) + \left(\frac{b}{\sqrt{a^2+b^2+c^2}} \right) \left(\frac{b}{\sqrt{a^2+b^2+c^2}} \right) + \left(\frac{c}{\sqrt{a^2+b^2+c^2}} \right) \left(\frac{c}{\sqrt{a^2+b^2+c^2}} \right)$$

$$\cos \alpha = \frac{-a^2+b^2+c^2}{a^2+b^2+c^2} \quad \text{--- (1)}$$

Let β be the angle between $\overrightarrow{AA'}$ and $\overrightarrow{BB'}$

$$\cos \beta = l_2 l_3 + m_2 m_3 + n_2 n_3$$

$$\cos \beta = \frac{-a^2-b^2+c^2}{a^2+b^2+c^2} \quad \text{--- (2)}$$

Let γ be the angle between $\overrightarrow{BB'}$ and $\overrightarrow{CC'}$

$$\cos \gamma = l_3 l_4 + m_3 m_4 + n_3 n_4$$

$$\cos \gamma = \frac{a^2-b^2-c^2}{a^2+b^2+c^2} \quad \text{--- (3)}$$

Let θ be the angle between $\overrightarrow{OO'}$ and $\overrightarrow{CC'}$

$$\cos \theta = l_1 l_4 + m_1 m_4 + n_1 n_4$$

$$\cos \theta = \frac{a^2+b^2-c^2}{a^2+b^2+c^2} \quad \text{--- (4)}$$

From (1),(2),(3) & (4) we see that the angles between four diagonals are

$$\cos(\text{angle}) = \frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2} \Rightarrow \text{angle} = \cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2} \right) \text{ hence proved.}$$

Q#16: A straight line makes angles of measure $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

Solution:

Let "a" be the length of each side of a cube

Points of the each corner of the cube are

$$O = (0,0,0), P = (a,a,a), A = (a,0,0), B = (0,a,0), C = (0,0,a)$$

$$A' = (0,a,a), B' = (a,0,a), C' = (a,a,0)$$

Now $\overrightarrow{OP}, \overrightarrow{AA'}, \overrightarrow{BB'}, \overrightarrow{CC'}$ are the diagonals of a cube

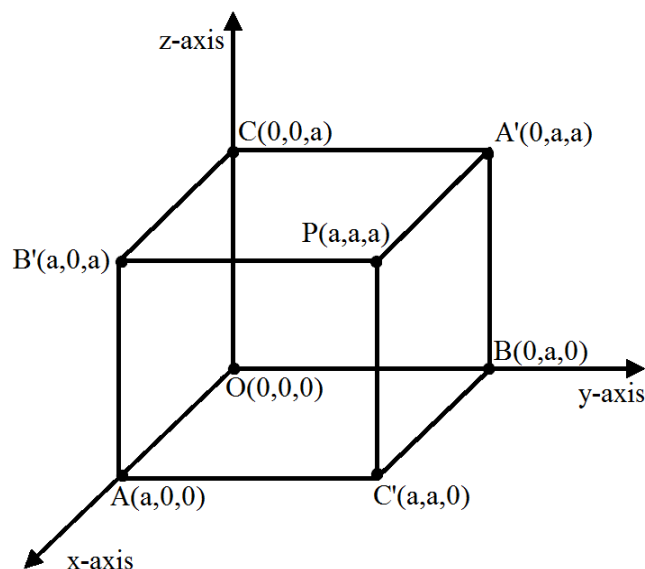
$$\overrightarrow{OP} = (a,a,a) - (0,0,0) = a\hat{i} + a\hat{j} + a\hat{k}$$

$$\overrightarrow{AA'} = (0,a,a) - (a,0,0) = -a\hat{i} + a\hat{j} + a\hat{k}$$

$$\overrightarrow{BB'} = (a,0,a) - (0,a,0) = a\hat{i} - a\hat{j} + a\hat{k}$$

$$\overrightarrow{CC'} = (a,a,0) - (0,0,a) = a\hat{i} + a\hat{j} - a\hat{k}$$

$$\text{Length of each diagonal is } \sqrt{a^2 + a^2 + a^2} = \sqrt{3a^2} = \sqrt{3}a$$



Now direction cosines of each diagonals are

$$\overrightarrow{OP} \Rightarrow l_1 = \frac{a}{\sqrt{3}a}, m_1 = \frac{a}{\sqrt{3}a}, n_1 = \frac{a}{\sqrt{3}a}$$

$$\overrightarrow{OP} \Rightarrow l_1 = \frac{1}{\sqrt{3}}, m_1 = \frac{1}{\sqrt{3}}, n_1 = \frac{1}{\sqrt{3}}$$

$$\overrightarrow{AA'} \Rightarrow l_2 = \frac{-1}{\sqrt{3}}, m_2 = \frac{1}{\sqrt{3}}, n_2 = \frac{1}{\sqrt{3}}$$

$$\overrightarrow{BB'} \Rightarrow l_3 = \frac{1}{\sqrt{3}}, m_3 = \frac{-1}{\sqrt{3}}, n_3 = \frac{1}{\sqrt{3}}$$

$$\overrightarrow{CC'} \Rightarrow l_4 = \frac{1}{\sqrt{3}}, m_4 = \frac{1}{\sqrt{3}}, n_4 = \frac{-1}{\sqrt{3}}$$

Let l, m, n be the direction cosines of line L which makes angles α, β, γ & δ with each diagonals

α is the angle between line L and $\overrightarrow{OP}$

$$\cos \alpha = ll_1 + mm_1 + nn_1 = l\left(\frac{1}{\sqrt{3}}\right) + m\left(\frac{1}{\sqrt{3}}\right) + n\left(\frac{1}{\sqrt{3}}\right) = \frac{l+m+n}{\sqrt{3}} \quad \text{--- (1)}$$

β is the angle between line L and $\overrightarrow{AA'}$

$$\cos \beta = ll_2 + mm_2 + nn_2 = l\left(\frac{-1}{\sqrt{3}}\right) + m\left(\frac{1}{\sqrt{3}}\right) + n\left(\frac{1}{\sqrt{3}}\right) = \frac{-l+m+n}{\sqrt{3}} \quad \text{--- (2)}$$

γ is the angle between line L and $\overrightarrow{BB'}$

$$\cos \gamma = ll_3 + mm_3 + nn_3 = l\left(\frac{1}{\sqrt{3}}\right) + m\left(\frac{-1}{\sqrt{3}}\right) + n\left(\frac{1}{\sqrt{3}}\right) = \frac{l-m+n}{\sqrt{3}} \quad \text{--- (3)}$$

δ is the angle between line L and $\overrightarrow{CC'}$

$$\cos \delta = ll_4 + mm_4 + nn_4 = l\left(\frac{1}{\sqrt{3}}\right) + m\left(\frac{1}{\sqrt{3}}\right) + n\left(\frac{-1}{\sqrt{3}}\right) = \frac{l+m-n}{\sqrt{3}} \quad \text{--- (4)}$$

Squaring eq. (1),(2),(3) & (4) and adding

$$\begin{aligned} \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta &= \left(\frac{l+m+n}{\sqrt{3}}\right)^2 + \left(\frac{-l+m+n}{\sqrt{3}}\right)^2 + \left(\frac{l-m+n}{\sqrt{3}}\right)^2 + \left(\frac{l+m-n}{\sqrt{3}}\right)^2 \\ &= \frac{l^2+m^2+n^2+2lm+2mn+2nl}{3} + \frac{l^2+m^2+n^2-2lm+2mn-2nl}{3} \\ &\quad + \frac{l^2+m^2+n^2-2lm-2mn+2nl}{3} + \frac{l^2+m^2+n^2+2lm-2mn-2nl}{3} \end{aligned}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4l^2 + 4m^2 + 4n^2}{3} = \frac{4(l^2 + m^2 + n^2)}{3}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3} \quad \text{hence proved}$$

Q#17: Find equations of the straight line passing through the point $P(0, -3, 2)$ and parallel to the straight line joining the points $A(3, 4, 7)$ and $B(2, 7, 5)$.

Solution: Consider two lines L_1 & L_2

Let L_1 be the required equation of the straight line passing through the point $P(0, -3, 2)$

The points $A(3, 4, 7)$ and $B(2, 7, 5)$ on line L_2 . So direction ratios from $A(3, 4, 7)$ to $B(2, 7, 5)$ are

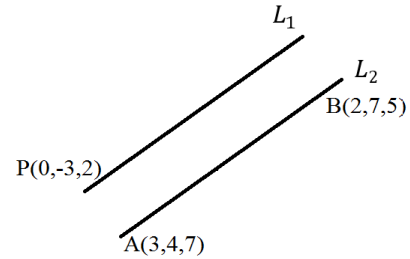
$$\overrightarrow{AB} = (2, 7, 5) - (3, 4, 7) = -\hat{i} + 3\hat{j} - 2\hat{k}$$

Then $-1, 3, -2$ are direction ratios of line L_2 .

By given condition both lines are parallel so both lines having same direction ratios.

Now required equation through the point $P(0, -3, 2)$

$$\frac{x-0}{-1} = \frac{y+3}{3} = \frac{z-2}{-2} \Rightarrow \frac{x}{1} = \frac{y+3}{-3} = \frac{z-2}{2} \quad \text{required equation.}$$



Q#18: Find equations of the straight line passing through the point $P(2, 0, -2)$ and perpendicular to each of straight lines

$$\frac{x-3}{2} = \frac{y}{2} = \frac{z+1}{2} \quad \text{and} \quad \frac{x}{3} = \frac{y+1}{-1} = \frac{z+2}{2}$$

Solution:

Given equations of lines are

$$\frac{x-3}{2} = \frac{y}{2} = \frac{z+1}{2} \quad \text{--- (L}_1\text{)}$$

$$\frac{x}{3} = \frac{y+1}{-1} = \frac{z+2}{2} \quad \text{--- (L}_2\text{)}$$

Direction ratios of line L_1 are $2, 2, 2 \Rightarrow \vec{d}_1 = 2\hat{i} + 2\hat{j} + 2\hat{k}$

Direction ratios of line L_2 are $3, -1, 2 \Rightarrow \vec{d}_2 = 3\hat{i} - \hat{j} + 2\hat{k}$

Suppose L be the required line through the point $P(2, 0, -2)$ with

direction ratios $c_1, c_2, c_3 \Rightarrow \vec{d}_3 = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

Since $L \perp L_1$

So by condition of perpendicularity

$$\vec{d}_1 \cdot \vec{d}_3 = 0 \Rightarrow (2\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) = 0$$

$$2c_1 + 2c_2 + 2c_3 = 0 \quad \text{----- (1)}$$

$$\vec{d}_2 \cdot \vec{d}_3 = 0 \Rightarrow (3\hat{i} - \hat{j} + 2\hat{k}) \cdot (c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) = 0$$

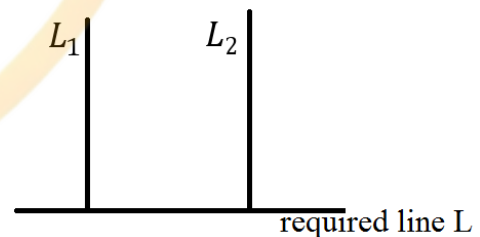
$$3c_1 - c_2 + 2c_3 = 0 \quad \text{----- (2)}$$

Now

$$\frac{c_1}{\begin{vmatrix} 2 & 2 \\ -1 & 2 \end{vmatrix}} = \frac{-c_2}{\begin{vmatrix} 2 & 2 \\ 3 & 2 \end{vmatrix}} = \frac{c_3}{\begin{vmatrix} 2 & 2 \\ 3 & -1 \end{vmatrix}} \Rightarrow \frac{c_1}{6} = \frac{-c_2}{-2} = \frac{c_3}{-8} \Rightarrow \frac{c_1}{3} = \frac{c_2}{1} = \frac{c_3}{-4}$$

Hence $(3, 1, -4)$ be the direction ratios of required line L

Now equation of required line L passing through the point $P(2, 0, -2)$ is $\frac{x-2}{3} = \frac{y-0}{1} = \frac{z+2}{-4}$



Find equations of straight line through the given point A and intersecting at right angles the given straight line:

Q#19: $A = (11, 4, -6)$ and $x = 4 - t, y = 7 + 2t, z = -1 + t$.

Solution: Let L be the required line passing through $A = (11, 4, -6)$ and perpendicular to given line.

Suppose it meets the given line at point B.

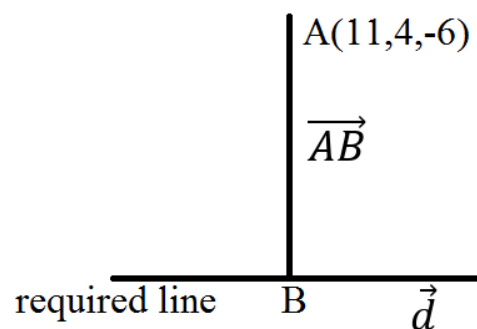
Now a point on given line is $B(4 - t, 7 + 2t, -1 + t)$

$$\overrightarrow{AB} = (4 - t - 11)\hat{i} + (7 + 2t - 4)\hat{j} + (-1 + t + 6)\hat{k}$$

$$\overrightarrow{AB} = (-t - 7)\hat{i} + (2t + 3)\hat{j} + (t + 5)\hat{k}$$

Hence direction vector of the given straight line is $\vec{d} = -\hat{i} + 2\hat{j} + \hat{k}$

Since $\overrightarrow{AB}$ is perpendicular to given line.



By perpendicular condition

$$\overrightarrow{AB} \perp \vec{d} = 0 \Rightarrow \overrightarrow{AB} \cdot \vec{d} = 0$$

$$[(-t - 7)\hat{i} + (2t + 3)\hat{j} + (t + 5)\hat{k}] \cdot [-\hat{i} + 2\hat{j} + \hat{k}] = 0$$

$$(-1)(-t - 7) + (2)(2t + 3) + (1)(t + 5) = 0 \Rightarrow 7 + t + 4t + 6 + t + 5 = 0 \Rightarrow t = -3$$

Direction vector of required line will become

$$\overrightarrow{AB} = (-(-3) - 7)\hat{i} + (2(-3) + 3)\hat{j} + (-3 + 5)\hat{k} \Rightarrow \overrightarrow{AB} = -4\hat{i} - 3\hat{j} + 2\hat{k}$$

Now required equation passing through the point $A = (11, 4, -6)$ having direction ratios $-4, -3, 2$

$$\frac{x-11}{-4} = \frac{y-4}{-3} = \frac{z+6}{2} \quad \text{OR} \quad \frac{x-11}{4} = \frac{y-4}{3} = \frac{z+6}{-2}$$

Q#20: $A = (5, -4, 4)$ and $\frac{x}{-1} = \frac{y-1}{1} = \frac{z}{-2}$

Solution: Given point and line are $A = (5, -4, 4)$ and $\frac{x}{-1} = \frac{y-1}{1} = \frac{z}{-2} = t$ (say)

The parametric equations of given line are $x = -t, y = 1 + t, z = -2t$

DO YOURSELF AS ABOVE

Q#21: Find the length of the perpendicular from the point $P(x_1, y_1, z_1)$ to the straight line

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}, \text{ where } l^2 + m^2 + n^2 = 1$$

Solution: Given point and line are

$$P(x_1, y_1, z_1) \quad \& \quad \frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$$

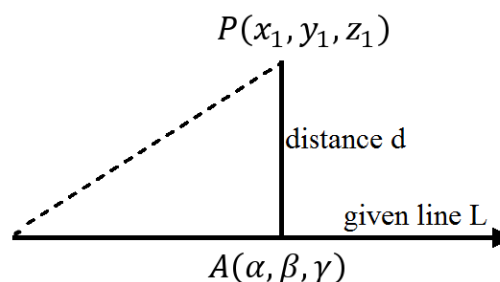
Hence $A = (\alpha, \beta, \gamma)$ is a point of given line

Direction vector of given line is $\vec{b} = l\hat{i} + m\hat{j} + n\hat{k}$

$$\text{Now } \overrightarrow{AP} = (x_1 - \alpha)\hat{i} + (y_1 - \beta)\hat{j} + (z_1 - \gamma)\hat{k}$$

Let d be the required distance then by using formula

$$d = \frac{|\overrightarrow{AP} \times \vec{b}|}{|\vec{b}|} \quad \text{--- (A)}$$



$$\begin{aligned}\therefore \overrightarrow{AP} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 - \alpha & y_1 - \beta & z_1 - \gamma \\ l & m & n \end{vmatrix} \\ &= [n(y_1 - \beta) - m(z_1 - \gamma)]\hat{i} - [n(x_1 - \alpha) - l(z_1 - \gamma)]\hat{j} + [m(x_1 - \alpha) - l(y_1 - \beta)]\hat{k} \\ \overrightarrow{AP} \times \vec{b} &= [n(y_1 - \beta) - m(z_1 - \gamma)]\hat{i} + [l(z_1 - \gamma) - n(x_1 - \alpha)]\hat{j} + [m(x_1 - \alpha) - l(y_1 - \beta)]\hat{k} \\ |\overrightarrow{AP} \times \vec{b}| &= \sqrt{[n(y_1 - \beta) - m(z_1 - \gamma)]^2 + [l(z_1 - \gamma) - n(x_1 - \alpha)]^2 + [m(x_1 - \alpha) - l(y_1 - \beta)]^2} \\ |\overrightarrow{AP} \times \vec{b}| &= \sqrt{\sum [n(y_1 - \beta) - m(z_1 - \gamma)]^2} \quad \& \quad |\vec{b}| = \sqrt{l^2 + m^2 + n^2} = \sqrt{1} = 1\end{aligned}$$

Putting in equation (A)

$$d = \frac{\sqrt{\sum [n(y_1 - \beta) - m(z_1 - \gamma)]^2}}{1} \Rightarrow d = \sqrt{\sum [n(y_1 - \beta) - m(z_1 - \gamma)]^2} \text{ required distance}$$

Q#22: Find equations of the perpendicular from the point P(1, 6, 3) to the straight line

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

Also obtain its length and coordinates of the foot of the perpendicular.

Solution:

Given point P(1,6,3) and equation of straight line is

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

Let AP be the length of perpendicular from point A to given line

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = t$$

So parametric equations of given line are

$$\begin{cases} x = t \\ y = 1 + 2t \\ z = 2 + 3t \end{cases}$$

Any point on this line is (t, 1 + 2t, 2 + 3t)

So coordinates of point P are P(t, 1 + 2t, 2 + 3t)

Direction ratios of line $\overrightarrow{AP}$ are = (t, 1 + 2t, 2 + 3t) - (1, 6, 3)

$$\overrightarrow{AP} = (t-1)\hat{i} + (2t-5)\hat{j} + (3t-1)\hat{k}$$

& direction vector of given line is $\vec{d} = \hat{i} + 2\hat{j} + 3\hat{k}$

$$\text{Since } \overrightarrow{AP} \perp \vec{d} \Rightarrow \overrightarrow{AP} \cdot \vec{d} = 0$$

$$[(t-1)\hat{i} + (2t-5)\hat{j} + (3t-1)\hat{k}] \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0$$

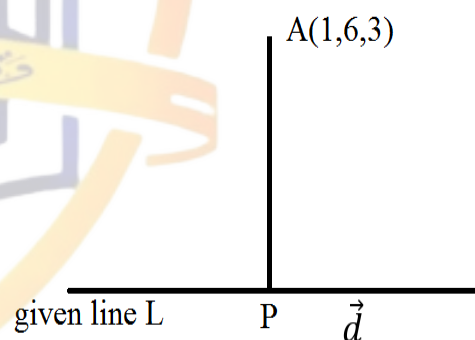
$$(1)(t-1) + (2)(2t-5) + (3)(3t-1) = 0 \Rightarrow 14t - 14 = 0 \Rightarrow t = 1$$

So coordinates of point P are P(1,3,5)

$$\text{Length of perpendicular} = |AP| = \sqrt{(1-1)^2 + (3-1)^2 + (5-3)^2} = \sqrt{0+9+4} = \sqrt{13}$$

Now equation of perpendicular $\overrightarrow{AP}$ is

$$\frac{x-1}{1-1} = \frac{y-6}{3-6} = \frac{z-3}{5-3} \Rightarrow \frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2} \text{ required equation of line}$$



Q#23: Find necessary and sufficient condition that the point $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$ and $R(x_3, y_3, z_3)$ are collinear.

Solution: Given points are $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$ & $R(x_3, y_3, z_3)$

Suppose that the points P, Q & R are collinear.

Now equation of line through $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$ is $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$

Since points P, Q & R are collinear. So point $R(x_3, y_3, z_3)$ lies on line.

$$\frac{x_3-x_1}{x_2-x_1} = \frac{y_3-y_1}{y_2-y_1} = \frac{z_3-z_1}{z_2-z_1} = t \text{ (say)}$$

$$\Rightarrow \begin{cases} x_3 - x_1 = t(x_2 - x_1) \\ y_3 - y_1 = t(y_2 - y_1) \\ z_3 - z_1 = t(z_2 - z_1) \end{cases} \quad \text{or} \quad \begin{cases} x_3 - x_1 - tx_2 + tx_1 = 0 \\ y_3 - y_1 - ty_2 + ty_1 = 0 \\ z_3 - z_1 - tz_2 + tz_1 = 0 \end{cases} \Rightarrow \begin{cases} (t-1)x_1 - tx_2 + x_3 = 0 \\ (t-1)y_1 - ty_2 + y_3 = 0 \\ (t-1)z_1 - tz_2 + z_3 = 0 \end{cases}$$

Eliminating $(t-1)$, $-t$, 1 from above equations

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0 \quad \text{Which is necessary condition for three points } P, Q \text{ & } R \text{ to be collinear.}$$

Q#24: If l_1, m_1, n_1 ; l_2, m_2, n_2 and l_3, m_3, n_3 are direction cosines of three mutual perpendicular lines, prove that the lines whose direction cosines are proportional to $l_1 + l_2 + l_3$, $m_1 + m_2 + m_3$, $n_1 + n_2 + n_3$ makes congruent angles with them.

Solution: Suppose that L_1, L_2 & L_3 are given line such that

Direction cosines of L_1 are l_1, m_1, n_1

Direction cosines of L_2 are l_2, m_2, n_2

Direction cosines of L_3 are l_3, m_3, n_3

Since lines L_1, L_2 & L_3 are mutually perpendicular

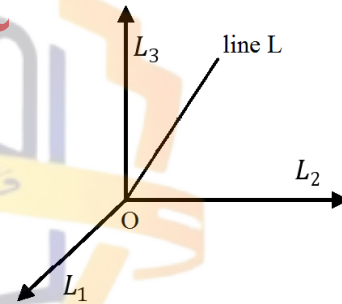
$$\text{So } \begin{cases} l_1 l_2 + m_1 m_2 + n_1 n_2 = 0 \\ l_2 l_3 + m_2 m_3 + n_2 n_3 = 0 \\ l_1 l_3 + m_1 m_3 + n_1 n_3 = 0 \end{cases}$$

Let L be the line having direction ratios $l_1 + l_2 + l_3$, $m_1 + m_2 + m_3$ & $n_1 + n_2 + n_3$

Let α be the angle between L & L_1 then

$$\begin{aligned} \cos \alpha &= \frac{l_1(l_1+l_2+l_3)+m_1(m_1+m_2+m_3)+n_1(n_1+n_2+n_3)}{\sqrt{l_1^2+m_1^2+n_1^2} \cdot \sqrt{(l_1+l_2+l_3)^2+(m_1+m_2+m_3)^2+(n_1+n_2+n_3)^2}} \\ &= \frac{(l_1^2+m_1^2+n_1^2)+(l_1 l_2+m_1 m_2+n_1 n_2)+(l_1 l_3+m_1 m_3+n_1 n_3)}{\sqrt{(l_1^2+m_1^2+n_1^2)+(l_2^2+m_2^2+n_2^2)+(l_3^2+m_3^2+n_3^2)+2(l_1 l_2+m_1 m_2+n_1 n_2)}} \end{aligned}$$

$$\cos \alpha = \frac{1+0+0}{\sqrt{1+1+1+0+0+0}} \Rightarrow \cos \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \text{ --- (1)}$$



Let β be the angle between L & L_2 then

$$\cos \beta = \frac{l_2(l_1 + l_2 + l_3) + m_2(m_1 + m_2 + m_3) + n_2(n_1 + n_2 + n_3)}{\sqrt{l_2^2 + m_2^2 + n_2^2} \cdot \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$= \frac{(l_1 l_2 + m_1 m_2 + n_1 n_2) + (l_2^2 + m_2^2 + n_2^2) + (l_2 l_3 + m_2 m_3 + n_2 n_3)}{\sqrt{(l_1^2 + m_1^2 + n_1^2) + (l_2^2 + m_2^2 + n_2^2) + (l_3^2 + m_3^2 + n_3^2) + 2(l_1 l_2 + m_1 m_2 + n_1 n_2)}}$$

$$\cos \beta = \frac{0+1+0}{\sqrt{1+1+1+0+0+0}} \Rightarrow \cos \beta = \frac{1}{\sqrt{3}} \Rightarrow \beta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \text{ --- (2)}$$

Let γ be the angle between L & L_1 then

$$\cos \gamma = \frac{l_3(l_1 + l_2 + l_3) + m_3(m_1 + m_2 + m_3) + n_3(n_1 + n_2 + n_3)}{\sqrt{l_3^2 + m_3^2 + n_3^2} \cdot \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$= \frac{(l_1 l_3 + m_1 m_3 + n_1 n_3) + (l_2 l_3 + m_2 m_3 + n_2 n_3) + (l_3^2 + m_3^2 + n_3^2)}{\sqrt{(l_1^2 + m_1^2 + n_1^2) + (l_2^2 + m_2^2 + n_2^2) + (l_3^2 + m_3^2 + n_3^2) + 2(l_1 l_2 + m_1 m_2 + n_1 n_2)}}$$

$$\cos \gamma = \frac{1+0+0}{\sqrt{1+1+1+0+0+0}} \Rightarrow \cos \gamma = \frac{1}{\sqrt{3}} \Rightarrow \gamma = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \text{ --- (3)}$$

From (1) (2) & (3) it is proved that $\Rightarrow \alpha \cong \beta \cong \gamma$ required result

Q#25: A variable line in two adjacent positions has direction cosines l, m, n and $l + \delta l, m + \delta m, n + \delta n$. Show that measure of the small angle $\delta\theta$ between the two positions is given by $(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$.

Solution:

Let OA and OB be the two adjacent positions of the line. Let PQ be the arc of the circle with centre at O and radius 1.

Then the coordinates of the points.

P & Q are $P(l, m, n)$ & $Q(l + \delta l, m + \delta m, n + \delta n)$.

Let $\delta\theta$ be the angle between two positions of line.

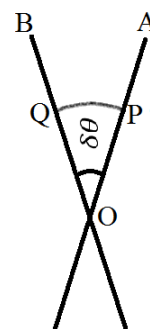
Now $\delta\theta = \text{chord PQ}$

So $\delta\theta = |PQ|$

$$\delta\theta = \sqrt{(l + \delta l - l)^2 + (m + \delta m - m)^2 + (n + \delta n - n)^2}$$

$$\delta\theta = \sqrt{\delta l^2 + \delta m^2 + \delta n^2}$$

$$\Rightarrow \delta\theta^2 = \delta l^2 + \delta m^2 + \delta n^2$$



Checked by: Sir Hameed ullah (hameedmath2017@gmail.com)

Specially thanks to my Respected Teachers

Prof. Muhammad Ali Malik (M.phill physics and Publisher of www.Houseofphy.blogspot.com)

Muhammad Umar Asghar sb (M.Sc Mathematics)

Hameed Ullah sb (M.Sc Mathematics)

